

COSM

Unit-1 : Probability: —

Sample space, Events, Counting ^{3am} points, Probability of an event, additive rules, conditional probability, independent events, product rule and Bayes theorem.

Random variables: Discrete and continuous random variables, expectation of random variables, moments, variance of random variables.

Random Experiment:- It is an experiment whose outcome cannot be predicted with certainty, even though all possible outcomes are known.

Ex:- Tossing a coin, Rolling a die, Drawing a card from a deck, measuring noise in a communication channel.

→ The Random experiment is also called a random trial.

Event:- A set of outcomes of a random experiment is called an event.

→ The outcomes of a random experiment are known as elementary events.

Ex:- when a coin is tossed, the events Head and Tail are mutually exclusive.

Probability:- when all outcomes of an experiment are equally likely, the probability of an event is defined as

probability of an event $(E) = P(E) =$

$\frac{\text{Number of favourable outcomes of } E}{\text{Total number of possible outcomes in the}}$

random experiment.

Ex:- when a fair die is thrown, the probability of getting an even number

$$\text{is } P(E) = \frac{3}{6} = \frac{1}{2}.$$

Since $E = \{2, 4, 6\}$, all possible outcomes are $\{1, 2, 3, 4, 5, 6\}$.

→ If $\bar{E}$ denotes the event of non-occurrence of E , then the probability of $\bar{E}$

$$\text{is } P(\bar{E}) = 1 - P(E).$$

$$\rightarrow P(E) + P(\bar{E}) = 1.$$

$$\rightarrow 0 \leq P(E) \leq 1.$$

→ If $P(E) = 1$, the event E is called certain event or sure event.

→ If $P(E) = 0$, the event E is called an impossible event.

Q. what is the probability for a leap year to have 52 Fridays and 53 Thursdays?

Sol:- A leap year has 366 days.

$$\text{and } 366 = 7 \times 52 + 2$$

$\therefore$ A leap year has 52 weeks and 2 extra days.

These 2 days can be any one of the following: (i) Mon, Tues (ii) Tues, wed, (iii) wed, Thurs (iv) Thurs, fri (v) fri, sat (vi) sat, sun (vii) Sun, Mon.

Let E be the event of a leap year having 52 Fridays and 53 Thursdays.

This happens only when the 2 extra days do not have Friday and have Thursday, i.e. (iii) case, wed, Thurs.

here no. of favourable ^{outcomes} events, $m = 1$

and total no. of outcomes, $n = 7$.

$$\therefore P(E) = \frac{m}{n} = \frac{1}{7}$$

Q. what is the probability that a leap year has only 52 Thursdays?

Ans:- $P(E) = \frac{5}{7}$

Q. Five digit numbers are formed with 0, 1, 2, 3, 4 (not allowing a digit being repeated in any number). Find the probability of getting 2 in the ten's place and 0 in the units place.

Sol: -

Total no. of 5 digit numbers using the digits 0, 1, 2, 3, 4 is $n = 4 \times 4 \times 3 \times 2 \times 1$
 $= 96$

Let E be the event of getting a number having 2 in 10's place and 0 in the units place.

$\therefore$ No. of favourable outcomes of E
 $= m = 3 \times 2 \times 1 = 6$

$$\therefore P(E) = \frac{m}{n} = \frac{6}{96} = \frac{1}{16} \quad \boxed{\underline{\quad} \underline{\quad} \underline{\quad} \underline{2} \underline{0}}$$

Q. A box contains n tickets marked 1 through n. Two tickets are drawn successively without replacement. Determine the probability that the no. on the tickets are consecutive integers.

Sol: - The box contains n tickets marked 1 through n.

The number of ways in which two tickets can be drawn without replacement = $n(n-1)$.

The total no. of favourable cases in which the numbers on the tickets are consecutive integers = $n-1$.

$\therefore$ Required probability = $\frac{n-1}{n(n-1)} = \frac{1}{n}$.

Simple Event - An event in a trial that cannot be further split is called a simple event or an elementary event.

Sample Space - The set of all possible simple events in a trial is called a

Sample space for the trial. It is denoted by S .

→ Each element of a sample space is called a sample point.

→ Any subset of a sample space is called an event. It is generally denoted by E .

→ A sample event is a sample point

→ ~~Sample space~~ is denoted by S .

Ex- Two coins are tossed.

$S = \text{sample space} = \{HH, HT, TH, TT\}$.

Simple events are HH, HT, TH, TT .
or (sample points)
Let E be an event of having 2 heads
or 2 tails, then $E = \{HH, TT\}$.

Then clearly $E \subset S$.

Mutually Exclusive Events: - Two events E_1, E_2 of a sample space S are said to be mutually exclusive if they have no common sample points. i.e. $E_1 \cap E_2 = \emptyset$.

Complementary Events: - Two mutually exclusive events E_1 and E_2 are said to be complementary events if $E_1 \cup E_2 = S$.

→ If E is an event of a sample space S , then its complement is denoted by

E' or $\bar{E}$.

$$\rightarrow P(E) + P(\bar{E}) = 1.$$

Axiomatic definition of probability:-
Let S be a finite sample space. A real valued function P from the powerset of S into R is called a probability function on S if the following three axioms are satisfied.

Axioms of Probability:-

- (i) Axiom of positivity:- $P(E) \geq 0$ for every subset E of S .
- (ii) Axiom of certainty:- $P(S) = 1$.
- (iii) Axiom of union:- If E_1, E_2 are disjoint subsets of S , then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

The image $P(E)$ of E is called the probability of the event E .

→ If $E_1, E_2, \dots, E_n$ are disjoint subsets of S , then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

Probability of an event in a sample space containing equally likely simple events.

Let a sample space S consists of n equally likely simple Events, $E_1, E_2, \dots, E_n$. Let $E = E_1 \cup E_2 \cup \dots \cup E_n$

The no. of sample points favourable to E
 $= n(E) = m.$

Let the Total no. of sample points in $S = n(S) = n.$

Now $P(E) = \frac{m}{n} = \frac{n(E)}{n(S)}.$

Q. In a group there are 3 men and 2 women. Three persons are selected at random from this group. Find the probability that one man and two women or two men and one woman are selected.

Sol: — Let S be the sample space of selecting 3 persons from 5 persons:

$$n(S) = \binom{5}{3} = 10$$

Let E_1 be the event of selecting 1 man and 2 women.

E_2 be the event of selecting 2 men and 1 woman.

No. of favourable outcomes: —

$$n(E_1) = \binom{3}{1} \binom{2}{2} = 3 \times 1 = 3$$

$$n(E_2) = \binom{3}{2} \binom{2}{1} = 3 \times 2 = 6$$

$$n(E_1 \cup E_2) = n(E_1) + n(E_2) \quad (\because E_1 \text{ and } E_2 \text{ are mutually exclusive}).$$

$$= 3 + 6 = 9.$$

$\therefore$ Required probability $= P(E_1 \cup E_2) =$

$$= \frac{n(E \cup E_2)}{n(S)} = \frac{9}{10}$$

Q. 12 balls are distributed at random among three boxes. what is the probability that the first box will contain 3 balls?

Sol: - Sample space: -

Each ball can go into any one of the 3 boxes.

$\therefore n(S) = \text{Total no. of possible}$

distributions = $3 \times 3 \times \dots \times 3$ (12 times)

$= 3^{12}$ (3 choices each)

Favourable outcomes: -

We want exactly 3 balls in the first box.

No. of ways in which 3 balls can go to the first box out of 12 balls = $\binom{12}{3}$

The remaining 9 balls can go into either box 2 or box 3 (2 choices each)

No. of distributions = $2 \times 2 \times \dots \times 2$ (9 times)

$$= 2^9$$

$\therefore \text{no. of favourable outcomes} = \binom{12}{3} \times 2^9$

Required probability: -

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{12}{3} 2^9}{3^{12}}$$

Q. out of 15 items 4 are not in good condition. 4 are selected at random. Find the probability that (i) All are not good (ii) Two are not good.

Sol Sample space —

$n(S) = \text{No. of ways of selecting 4 items from 15}$
 $= \binom{15}{4}$

Favourable outcomes: —

(i) Let E_1 be the event of selecting 4, which are not good.

$$n(E_1) = 4$$

$$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{4}{\binom{15}{4}} = \frac{4}{15! \times 11! 4!} = \frac{1}{1365}$$

(ii) Let E_2 be the event of selecting 2 items, not in good condition.

$$n(E_2) = \binom{4}{2}$$

$$\& P(E_2) = \frac{\binom{4}{2}}{\binom{15}{4}} = \frac{2}{455}$$

Addition theorem on Probability: - If S is a sample space and E_1, E_2 are any events in S then $P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

Proof: - Case (i): - $E_1 \cap E_2 \neq \emptyset$

Let $E_1 = \{a_1, a_2, \dots, a_k, a_{k+1}, a_{k+2}, \dots, a_{k+l}\}$
 $E_2 = \{a_{k+1}, a_{k+2}, \dots, a_{k+l}, a_{k+l+1}, \dots, a_{k+l+m}\}$

$E_1 \cup E_2 = \{a_1, a_2, \dots, a_k, a_{k+1}, a_{k+2}, \dots, a_{k+l}, a_{k+l+1}, \dots, a_{k+l+m}\}$

(ii) $E_1 \cap E_2 = \{a_{k+1}, a_{k+2}, \dots, a_{k+l}\}$

$$\begin{aligned} \therefore P(E_1) + P(E_2) - P(E_1 \cap E_2) &= P(a_1) + P(a_2) + \dots + P(a_k) + P(a_{k+1}) + P(a_{k+2}) \\ &+ \dots + P(a_{k+l}) + P(a_{k+l+1}) + P(a_{k+l+2}) + \dots + P(a_{k+l+m}) \\ &- P(a_{k+1}) - P(a_{k+2}) - \dots - P(a_{k+l}) \\ &= P(a_1) + P(a_2) + \dots + P(a_k) + P(a_{k+1}) + \dots + P(a_{k+l+m}) \\ &= P(E_1 \cup E_2) \end{aligned}$$

Case (ii) $E_1 \cap E_2 = \emptyset$

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) \\ &= P(E_1) + P(E_2) - 0 \\ &= P(E_1) + P(E_2) - P(\emptyset) \\ &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \end{aligned}$$

hence proved.

Q. A card is drawn from a well-shuffled pack of cards. what is the probability that it is either a spade or an ace?

Soln. Let S be the sample space, of drawing a card from a well shuffled pack, then $n(S) = 52$.

Let A denote the event of getting a spade and B be the event of getting an ace.

Then $A \cup B$ is the event of getting a spade or an ace.

$A \cap B$ is the event of getting a spade and an ace.

$$P(A) = \frac{13}{52}, P(B) = \frac{4}{52}, P(A \cap B) = \frac{1}{52}$$

$\therefore$ By addition theorem,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13} \therefore$$

Q. Three students A, B, C are in running race. A and B have the same probability of winning and each is twice as likely to win as C. find the probability that B or C wins.

Sol: — Sample space of race, $S = A \cup B \cup C$.

Given, $P(A) = P(B)$ and $P(A) = 2P(C)$,

$$P(B) = 2P(C). \quad \text{--- (1)}$$

ekkt, $P(A) + P(B) + P(C) = 1$ ($\because P(S) = 1$)

$$\rightarrow 2P(C) + 2P(C) + P(C) = 1$$

$$\rightarrow 5P(C) = 1 \rightarrow P(C) = \frac{1}{5}. \quad \text{--- (2)}$$

from (1), & (2), $P(A) = \frac{2}{5}$, $P(B) = \frac{2}{5}$

Now the probability that B or C wins

$$= P(B \cup C) = P(B) + P(C) + P(B \cap C)$$

$$= \frac{2}{5} + \frac{1}{5} + 0 = \frac{3}{5}$$

Conditional event: — If A and B are two events, then "Event A given that B has occurred" is called a conditional event and is denoted by $A|B$.

Conditional Probability: —

The probability of the conditional event $A|B$ is $P(A|B) = \frac{P(A \cap B)}{P(B)}$ when $P(B) > 0$.

Similarly, $P(B|A) = \frac{P(A \cap B)}{P(A)}$ when $P(A) > 0$.

Multiplication Theorem of Probability: —

In a random experiment, if A, B are two events such that $P(A) \neq 0$ and $P(B) \neq 0$, then $P(A \cap B) = P(B) \cdot P(A|B) = P(A) \cdot P(B|A)$.

Proof:- Let S be a sample space associated with a random experiment.

Let A and B be two events of S such that $P(A) \neq 0$, $P(B) \neq 0$.

$\therefore P(A) \neq 0$, by definition of Conditional probability, $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$$\rightarrow P(A \cap B) = P(A) \cdot P(B|A)$$

Similarly $P(A \cap B) = P(B) \cdot P(A|B)$ ($\because P(B) \neq 0$)

$\rightarrow$ If A and B are two events ($\neq \emptyset$),

then $P(\bar{A} \cup \bar{B}) = 1 - P(A \cap B) = 1 - P(A) \cdot P(B|A)$

$\rightarrow$ If A and B are independent events, i.e. the occurrence of the event A is not affected by the occurrence or non-occurrence of the event B , then the event A is said

$$P(A|B) = P(A)$$

$\rightarrow$ If A and B are dependent

$$\text{then } P(A|B) \neq P(A)$$

$$\rightarrow P(A \cap B \cap C) = P(A) \cdot P(A|B) \cdot P(C|A \cap B)$$

Q. A class has 10 boys and 5 girls. Three students are selected at random one after another without replacement. Find the probability that (i) the first two selected are boys, and the third is a girl. (ii) The first and third selected are boys and the second is a girl.

Sol Total no. of students =

$$10 \text{ boys} + 5 \text{ girls} = 15$$

(i) The probability that first two are boys and the third is girl is

$$P(E) = \frac{10}{15} \times \frac{9}{14} \times \frac{5}{13} = \frac{15}{91}$$

(ii). To find the probability that the first and third are boys and second is a girl

$$P(E) = \frac{10}{15} \times \frac{5}{14} \times \frac{9}{13} = \frac{15}{91}$$

Q. Two aeroplanes bomb a target successively. The probability of each correctly scoring a hit is 0.3 and 0.2 respectively. The second will bomb only if the first missed the target. Find the

probability that (i) target is hit
(ii) both fails to score hits.

Sol Let A be the event of first plane hitting the target.

Let B be the event of 2nd plane hitting the target.

Given $P(A) = 0.3$, $P(B|A) = 0.2$.

(i) Probability that target is hit:-

Target can be hit in 2 mutually exclusive ways:-

(i) 1st plane hits

(ii) 1st misses & 2nd hits.

$\therefore$ Required probability $= P(A) + P(\bar{A} \cap B)$

$$= P(A) + P(B|\bar{A}) \cdot P(\bar{A})$$

$$= 0.3 + 0.2 \cdot (1 - 0.3)$$

$$= 0.3 + 0.2(0.7) = 0.3 + 0.14 = 0.44$$

(ii) Probability that both fails to score hits:-

$$= P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}|\bar{A})$$

$$= (1 - 0.3) \cdot (1 - 0.2)$$

$$= (0.7)(0.8) = 0.56$$

Q. Determine (i) $P(\frac{B}{A})$ (ii) $P(\frac{A}{B^c})$ if A and B are events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cap B) = \frac{1}{12}$.

Solⁿ (i) wkt $\boxed{P(\frac{B}{A}) = \frac{P(A \cap B)}{P(A)}}$

Given $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cup B) = \frac{1}{2}$

By addition theorem, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\rightarrow \frac{1}{2} = \frac{1}{3} + \frac{1}{4} - P(A \cap B)$$

$$\rightarrow P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{2} = \frac{4+3-6}{12} = \frac{1}{12}$$

$$\therefore \text{(i)} P(\frac{B}{A}) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{3}{12} = \frac{1}{4}$$

(ii) wkt $P(\frac{A}{B^c}) = \frac{P(A \cap B^c)}{P(B^c)}$

& wkt $P(A \cap B^c) = P(A) - P(A \cap B)$

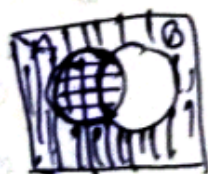
$$= \frac{1}{3} - \frac{1}{12} = \frac{4-1}{12} = \frac{3}{12}$$

$$= \frac{1}{4}$$

$$\& P(B^c) = 1 - P(B) = 1 - \frac{1}{4} = \frac{4-1}{4} = \frac{3}{4}$$

$$\therefore P(\frac{A}{B^c}) = \frac{P(A \cap B^c)}{P(B^c)}$$

$$= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$$



Q. Box A contains 5 red and 3 white marbles and box B contains 2 red and 6 white marbles. If a marble is drawn from each box, what is the probability that they are both of same colour.

Sol:- Let E_1 be the event that the marble is drawn from box A is Red.

Let E_2 be the event that the marble is drawn from box B is Red.

$$P(E_1) = \frac{1}{2} \cdot \frac{5}{8} = \frac{5}{16}$$

Box A: 5R & 3W

$$P(E_2) = \frac{1}{2} \cdot \frac{2}{8} = \frac{1}{8}$$

Box B: 2R & 6W

The probability that both the marbles are red is $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

$$= \frac{5}{16} \cdot \frac{1}{8} = \frac{5}{128}$$

Let E_3 be the event that the marble is drawn from box A and is white.

Let E_4 be the event that the marble is drawn from box B and is white.

$$P(E_3) = \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{16}$$

$$P(E_4) = \frac{1}{2} \cdot \frac{6}{8} = \frac{3}{8}$$

$\therefore$ The probability that both the marbles are white is $P(E_3 \cap E_4) = P(E_3) \cdot P(E_4)$

$$= \frac{3}{16} \cdot \frac{3}{8} = \frac{9}{128}$$

hence the probability that the marbles are of same colour = $\frac{5}{128} + \frac{9}{128} = \frac{14}{128} = \frac{7}{64}$

Q. Two marbles are drawn from a box containing 10 red, 30 white, 20 blue and 15 orange marbles, with replacement being made after each draw. Find the probability that (i) Both are white.
(ii) first is red and second is white.

Sol: - (i) Probability that both are white
$$= \frac{30}{75} \cdot \frac{30}{75}$$

(ii) Probability that first is red and second is white

$$= \frac{10}{75} \cdot \frac{30}{75}$$

Two cards are drawn from a well shuffled pack of 52 cards. Find the probability that they are both aces if the first card is
(i) replaced (ii) not replaced.

Sol: Using permutations:—

Total no. of cards = 52

no. of cards drawn = 2

no. of Ace cards = 4

(i) no. of total outcomes when the

first card is replaced = 52×52

no. of ways of drawing 2 aces = 4×4

$\therefore$ Required probability = $\frac{4 \times 4}{52 \times 52} = \frac{1}{169}$

(ii) no. of ways of drawing

2 cards without replacement = 52×51

no. of ways of drawing 2 aces = 4×3

$\therefore$ Required probability = $\frac{4 \times 3}{52 \times 51} = \frac{1}{221}$

Using Combinations:—

(i) no. of ways of drawing 2 from 52 = ${}^{52}C_2$

no. of ways of drawing 2 aces = 4C_2

$\therefore$ Required probability = $\frac{{}^4C_2}{{}^{52}C_2}$

$$= \frac{4!}{(4-2)!2!} \div \frac{52!}{(52-2)!2!} = \frac{4!}{2!2!} \times \frac{50!}{52!}$$

$$= \frac{4 \times 3 \times 50!}{52 \times 51 \times 50!} = \frac{4 \times 3}{52 \times 51} = \frac{1}{221}$$

Using Conditional probability:-
Let A be an event of drawing first card
an Ace.

B: second card drawn is an Ace.

To find $P(A \cap B) = P(A) \cdot P(B|A)$

(i) with Replacement

$$P(A) = \frac{4}{52}, \quad P(B|A) = \frac{4}{52}$$

$$P(A \cap B) = \frac{4}{52} \cdot \frac{4}{52} = \frac{1}{169}$$

(ii) without Replacement:-

$$P(A) = \frac{4}{52}, \quad P(B|A) = \frac{3}{51}$$

$$P(A \cap B) = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$$

Baye's theorem:- $E_1, E_2, \dots, E_n$ are n
mutually exclusive and exhaustive events
such that $P(E_i) > 0$ ($i=1, 2, \dots, n$) in a
sample space S and A is any other event
in S intersecting with every E_i such that
 $P(A) > 0$. Then for any event E_k ,

$$P(E_k|A) = \frac{P(E_k) \cdot P(A|E_k)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \dots + P(E_n)P(A|E_n)}$$

Proof:- Given $E_1, E_2, \dots, E_n$ are
 n events in S such that $P(E_i) > 0$

$$E_i \cap E_j = \emptyset \text{ for } i \neq j$$

$$S = E_1 \cup E_2 \cup \dots \cup E_n$$

also given that A is an event in S with $P(A) > 0$ & $P(E_i \cap A) \neq \emptyset$.

$$\text{then } A = A \cap S \\ = A \cap (E_1 \cup E_2 \cup \dots \cup E_n)$$

$$A = (A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)$$

note that $A \cap E_1, A \cap E_2, \dots, A \cap E_n$ are mutually exclusive.

$$\text{Then } P(E_k | A) = \frac{P(A \cap E_k)}{P(A)} \\ = \frac{P(A \cap E_k)}{P[(A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)]}$$

$$= \frac{P(A \cap E_k) \cdot P(E_k)}{P(A \cap E_1) \cdot P(E_1) + P(A \cap E_2) \cdot P(E_2) + \dots + P(A \cap E_n) \cdot P(E_n)} \quad \text{from (1)}$$

Q. of the 3 men, the chances that a politician, a business man or an academican, will be appointed as a vice-chancellor (v.c) of a university are 0.5, 0.3, 0.2 respectively.

Probability that research is promoted by these persons if they are appointed as v.c are 0.3, 0.7, 0.8 respectively.

(i) Determine the probability that research is promoted.

(ii) If research is promoted, what is the probability that v.c. is an academician?

Sol:— Let A, B, C be the events that a politician, a business man or an academician are appointed as v.c. respectively.

given that $P(A) = 0.5$

$P(B) = 0.3$

$P(C) = 0.2$

Let R be the event of promoting research

given that $P(R|A) = 0.3$

$P(R|B) = 0.7$

$P(R|C) = 0.8$

(i) The probability that research is promoted:—

$$P(R) = P(A)P(R|A) + P(B) \cdot P(R|B) + P(C) \cdot P(R|C)$$

$$= (0.5)(0.3) + (0.3)(0.7) + (0.2)(0.8)$$

$$= 0.52$$

(ii) The probability that research is promoted when the v.c. is an academician assuming that research is promoted.

$$= P(C|R) = \frac{P(R|C) \cdot P(C)}{P(R|A) \cdot P(A) + P(R|B) \cdot P(B) + P(R|C) \cdot P(C)}$$

$$= \frac{0.8 \times 0.2}{0.5 \times 0.3 + 0.3 \times 0.7 + 0.2 \times 0.8}$$

$$= \frac{4}{13} = 0.30769$$

Q. In a bolt factory, machines A, B, C manufacture 20%, 30% and 50% of the total of their output and 6%, 3% and 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from (i) Machine A (ii) Machine B (iii) Machine C.

Sol. Let A, B, C be the events that the bolts are manufactured by the machines A, B, C respectively.

$$\text{then } P(A) = \frac{20}{100}, P(B) = \frac{30}{100},$$

$$P(C) = \frac{50}{100}$$

$$\text{ie. } \frac{1}{5}, \frac{3}{10}, \frac{1}{2}$$

Let D be the event that the bolt manufactured is defective.

$$\text{Given that } P(D|A) = \frac{6}{100}, P(D|B) = \frac{3}{100}$$

$$P(D|C) = \frac{2}{100}$$

(i) If bolt is defective, then the probability that it is from machine-A

$$= P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B) + P(D|C) \cdot P(C)}$$

$$= \frac{12}{31}$$

$$= \frac{12}{31}$$

Similarly

$$(ii) P(B|D) = \frac{P(D|B) \cdot P(B)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B) + P(D|C) \cdot P(C)}$$

$$= \frac{9}{31}$$

$$= \frac{9}{31}$$

$$(iii) P(C|D) = \frac{10}{31}$$

Q. A businessman goes to hotels X, Y, Z, 20%, 50%, 30% of the time respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbing, what is the probability that business man's room having faulty plumbing is assigned to hotel Z?

Soln Given that the probabilities of a business man going to hotels X, Y, Z hotels are $P(X) = \frac{20}{100}$, $P(Y) = \frac{50}{100}$, $P(Z) = \frac{30}{100}$

$$P(Z) = \frac{30}{100}$$

$$\text{i.e. } P(X) = \frac{2}{10}, P(Y) = \frac{5}{10}, P(Z) = \frac{3}{10}$$

Let E be an event of having faulty plumbing.

$$\text{Given that } P(E|X) = \frac{5}{100} = \frac{1}{20}$$

$$P(E|Y) = \frac{4}{100} = \frac{1}{25}$$

$$P(E|Z) = \frac{8}{100} = \frac{2}{25}$$

$$\text{To find } P(Z/E) = \frac{P(E|Z) \cdot P(Z)}{P(E|X) \cdot P(X) + P(E|Y) \cdot P(Y) + P(E|Z) \cdot P(Z)}$$

$$= \frac{\frac{2}{25} \cdot \frac{3}{10}}{\frac{1}{20} \cdot \frac{2}{10} + \frac{1}{25} \cdot \frac{5}{10} + \frac{2}{25} \cdot \frac{3}{10}}$$

$$= \frac{\frac{6}{250}}{\frac{1}{10} + \frac{1}{10} + \frac{6}{250}}$$

$$= \frac{\frac{6}{250}}{\frac{2}{10} + \frac{6}{250}} = \frac{\frac{6}{250}}{\frac{100}{125} + \frac{6}{250}} = \frac{\frac{6}{250}}{\frac{126}{125}} = \frac{6}{126} = \frac{1}{21}$$

$$= \frac{\frac{6}{250}}{\frac{2}{10} + \frac{1}{10} + \frac{6}{250}} = \frac{\frac{6}{250}}{\frac{100}{125} + \frac{6}{250}} = \frac{\frac{6}{250}}{\frac{126}{125}} = \frac{6}{126} = \frac{1}{21}$$

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Random Variables and Distribution Functions

Random variables: It is a function that assigns a real number to each outcome of a random experiment.

Ex: Tossing a coin

outcome = Head or Tail.

define random variable x : $x=1$ if H
 $x=0$ if Tail.

Types of Random variable —

(i). Discrete random variables: takes countable values.

(ii) Continuous random variable: takes values over an interval.

(*) Expectation of a Discrete Random Variable

Let x be a discrete random variable, taking values $x_1, x_2, \dots, x_n$ with probabilities $p(x=x_i) = p_i$, then $E(x) = \sum_{i=1}^n x_i p_i$ is said to be the mathematical expectation or mean or expected value of x .

Properties $\rightarrow \boxed{\sum p_i = 1}$

$\rightarrow E(c) = c$, constant.

$$\rightarrow E(ax+b) = a E(X) + b.$$

$$\rightarrow E(X+Y) = E(X) + E(Y).$$

$\rightarrow$ Mean is another name of expectation, $\mu = E(X)$.

2) Variance: - It measures the spread or dispersion of a random variable about its mean.

$$\text{Var}(X) = E[(X - \mu)^2] \text{ where } \mu = E(X)$$

$$\text{or } \text{Var}(X) = E(X^2) - [E(X)]^2$$

✓ Variance for Discrete Random Variable

$$\text{Var}(X) = \sum (x_i - \mu)^2 p_i = (\sum x_i^2 p_i) - \mu^2$$

Properties:

$$\rightarrow \text{Var}(c) = 0$$

$$\rightarrow \text{Var}(aX) = a^2 \text{Var}(X)$$

$$\rightarrow \text{Var}(X+c) = \text{Var}(X).$$

(σ) Standard deviation: - It is the positive square root of variance.

$$\sigma = \sqrt{\text{Var}(X)}$$

$\rightarrow$ variance is in squared units,

where as S.D. is in original units.

$$\text{Unit of variance} = (\text{Unit of } X)^2$$

$$\text{Unit of (S.D.)} = \text{Unit of } X$$

Small variance: - Values are tightly clustered.

Large variance: - Values are widely spread.

$\rightarrow$ SD is less $\leftarrow$ Data values are close to the mean
low variability / low fluctuation
More consistent or stable data

Result If x is a discrete random variable, then $v(ax+b) = a^2 v(x)$ where $v(x)$ is variance of x and a, b are constants.

Proof Let $Y = ax + b$ — (1)

then $E(Y) = E(ax+b) = aE(x) + b$ — (2)

(1)-(2) $\Rightarrow Y - E(Y) = ax + b - aE(x) - b$
 $= a(x - E(x))$

squaring and taking expectation of both sides, we get

$$E(Y - E(Y))^2 = a^2 E(x - E(x))^2$$
$$v(Y) = a^2 v(x)$$

Discrete Probability distribution (Probability mass function): — A DPD assigns probabilities to each possible value of a discrete random variable such that the sum of all probabilities is equal to one.

Def: Let x be a discrete Random variable. Then the function $P(X=x) = P(x)$ is called the probability mass function of x .

properties: 1. $P(x) \geq 0 \quad \forall x$

2. $\sum P(x) = 1$

3. probability of any event is obtained by adding probabilities of all values of the random variable that enter the event.

Ex- Tossing a fair coin

x	0	1
$P(x)$	$1/2$	$1/2$

Ex- A random variable x has the following distribution:

x	1	2	3
$P(x)$	0.2	0.5	0.3

To verify that it is a valid distribution:
 $\sum P(x) = 0.2 + 0.5 + 0.3 = 1$, yes, valid.

To find probability: $P(x \geq 2)$

$$P(x \geq 2) = P(2) + P(3) = 0.5 + 0.3 = 0.8.$$

Q. Two dice are thrown. Let x be assigned to each point (a, b) in S the maximum of its numbers i.e. $x(a, b) = \max(a, b)$.

Find the probability distribution. x is a random variable with $x(s) = \{1, 2, 3, 4, 5, 6\}$.

Also find the mean and variance of the distribution.

Sol- The total no. of outcomes when 2 dice are thrown is $6 \times 6 = 36$.

$$S = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$$

Given that $x(a, b) = \max(a, b)$.

$$X(1,1) = 1$$

$$X(1,2), X(2,1), X(2,2) = 2$$

$$X(1,3), X(3,1), X(2,3), X(3,2), X(3,3) = 3$$

$$X(1,4), X(4,1), X(2,4), X(4,2), X(4,3), X(3,4), X(4,4) = 4$$

$$X(1,5), X(5,1), X(2,5), X(5,2), X(3,5), X(5,3), X(4,5), X(5,4), X(5,5) = 5$$

$$X(1,6), X(6,1), X(6,2), X(2,6), X(3,6), X(6,3), X(4,6), X(6,4), X(5,6), X(6,5), X(6,6) = 6$$

$$\therefore P(X=1) = \frac{1}{36} \quad P(X=4) = \frac{7}{36}$$

$$P(X=2) = \frac{3}{36} \quad P(X=5) = \frac{9}{36}$$

$$P(X=3) = \frac{5}{36} \quad P(X=6) = \frac{11}{36}$$

$\therefore$ The Required probability distribution is

$X = x_i$	1	2	3	4	5	6
$P(X = x_i) = p(x_i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$(i) \text{ Mean, } \mu = \sum_{i=1}^6 x_i p(x_i) = 1 \cdot \frac{1}{36} + 2 \cdot \frac{3}{36} + 3 \cdot \frac{5}{36} + 4 \cdot \frac{7}{36} + 5 \cdot \frac{9}{36} + 6 \cdot \frac{11}{36} = \frac{161}{36} = 4.47$$

$$(ii) \text{ Variance, } \sigma^2 = \sum_{i=1}^6 (x_i - \mu)^2 p(x_i)$$

$$\sum_{i=1}^6 x_i^2 p(x_i) - \mu^2 = 21.97 - 19.981 = 1.9912$$

Q. A random variable x has the following probability function:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) Determine k . (ii) Evaluate $P(X < 6)$,

$P(X \geq 6)$, $P(0 < X < 5)$ and $P(0 \leq X \leq 4)$.

(iii) If $P(X \leq k) > \frac{1}{2}$, find the minimum value of k and (iv) Determine the

distribution function of x (v) Mean

(vi) variance.

Sol: $\sum_{i=0}^7 p(x_i) = 1$

$$(i) \therefore 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$8k + 10k^2 + k = 1$$

$$\rightarrow 10k^2 + 9k - 1 = 0$$

$$\rightarrow 10k^2 + 10k - k - 1 = 0$$

$$\rightarrow 10k(k+1) - 1(k+1) = 0$$

$$\rightarrow (k+1)(10k-1) = 0$$

$$\rightarrow k = -1; \frac{1}{10}$$

$\therefore$ probability can not be negative

$$\boxed{k = \frac{1}{10}}$$

$$(ii) P(X < 6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= k^2 + 8k$$

$$= \left(\frac{1}{10}\right)^2 + \frac{8}{10} = \frac{1}{100} + \frac{8}{10}$$

$$= \frac{81}{100} = 0.81$$

$$\begin{aligned}
 P(X \geq 6) &= P(X=6) + P(X=7) \\
 &= 2k^2 + 7k^2 + k = 9k^2 + k = 9\left(\frac{1}{10}\right)^2 + \frac{1}{10} \\
 &= \frac{9}{100} + \frac{1}{10} = \frac{19}{100} = 0.19
 \end{aligned}$$

$$\begin{aligned}
 P(0 < X < 5) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\
 &= k + 2k + 2k + 3k = 8k = \frac{8}{10} = 0.8
 \end{aligned}$$

$$\begin{aligned}
 P(0 \leq X \leq 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\
 &\quad + P(X=4) = 8k = 0.8
 \end{aligned}$$

(iii) Given $P(X \leq k) > \frac{1}{2} = 0.5$

$$\begin{aligned}
 P(X \leq 1) &= P(X=0) + P(X=1) \\
 &= 0 + k = k = \frac{1}{10} = 0.1 < \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\
 &= 0 + k + 2k = 3k = \frac{3}{10} = 0.3 < 0.5
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\
 &= 0 + k + 2k + 2k = 5k = \frac{5}{10} = 0.5 = 0.5
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\
 &\quad + P(X=4) = 8k = \frac{8}{10} = 0.8 > 0.5 = \frac{1}{2}
 \end{aligned}$$

$\therefore 4$ is the minimum value of k

such that $P(X \leq k) > \frac{1}{2}$

$$\text{(iv) Mean} = \sum x_i p_i = 0 + k + 4k + 8k +$$

$$12k + 5k^2 + 6(2k^2) + 7(7k^2 + k) +$$

$$= 23k + 5k^2 + 12k^2 + 49k^2 + 7k$$

$$= 66k^2 + 30k = \frac{66}{100} + \frac{30}{10} = 3.66$$

$$\therefore \mu = 3.66$$

$$(v). \text{ Variance } = \sigma^2 = \sum_{i=0}^n x_i^2 p_i - \mu^2$$

$$K + 8K + 18K + 48K + 25K^2 + 72K^2 + 343K^2 + 49K - (3.66)^2$$

$$= 440K^2 + 124K - 3.66^2 = \frac{440}{100} + \frac{124}{10} - (3.66)^2$$

$$= 4.4 + 12.4 - 13.3956 = 3.4044$$

Q. The probability density function of a variate x is

x	0	1	2	3	4	5	6
$P(x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find k (ii) Find $P(x < 4)$, $P(x \geq 5)$, $P(3 < x \leq 6)$

(iii). what will be the minimum value of k so that $P(x \leq 2) > 0.3$?

Sol - (i) $k = \frac{1}{49}$ (ii). $P(x < 4) = \frac{16}{49}$,

$$P(x \geq 5) = \frac{24}{49}; \quad P(3 < x \leq 6) = \frac{33}{49}$$

(iii). $P(x \leq 2) = 9k \neq \frac{9}{49} = 0.18 \text{ } k \neq \frac{1}{30}$

Given $P(x \leq 2) > 0.3$.

$$9k > 0.3$$

$$k > \frac{0.3}{9}$$

$$\text{ie. } k > \frac{0.1}{3}$$

$$\text{ie. } k > \frac{1}{30}$$

$\therefore$ The minimum value of k is $\frac{1}{30}$ so that $P(x \leq 2) > 0.3$.

Q. From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn at random. Let the random variable X denote the no. of defective items in the sample. Find the probability distribution of X when the sample is drawn without replacement.

Sol: — Drawing is a sample without replacement is a Random Experiment

$\therefore$ the no. of defective items = 3

X is the no. of defective items in the sample, X takes either 0, 1, 2 or

No. of Good items = 7.

$$P(X=0) = P(\text{No defective}) = \frac{{}^7C_4}{{}^{10}C_4} = \frac{1}{252}$$

$$P(X=1) = P(3 \text{ Good \& } 1 \text{ defective}) \\ = \frac{{}^7C_3 \times {}^3C_1}{{}^{10}C_4} = \frac{1}{21}$$

$$P(X=2) = P(2 \text{ Good \& } 2 \text{ defective}) \\ = \frac{{}^7C_2 \times {}^3C_2}{{}^{10}C_4} = \frac{3}{105}$$

$$P(x=3) = P(\text{all defective \& good})$$

$$= {}^7C_1 \times {}^3C_3 = \frac{1}{30}$$

The probability distribution of random variable x is as follows:

x	0	1	2	3
$p(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

Q. Find the mean and variance of the uniform probability distribution given by $f(x) = \frac{1}{n}$ for $x = 1, 2, 3, \dots, n$.

Sol: - The probability distribution is

x	1	2	3	...	n
$f(x)$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	...	$\frac{1}{n}$

$$(i) \text{ Mean} = \sum_{i=1}^n x_i f(x_i) = 1 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} + \dots + n \cdot \frac{1}{n}$$

$$= \frac{1}{n} (1 + 2 + \dots + n) = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$(ii) \text{ Variance} = \sum_{i=1}^n x_i^2 f(x_i) - \mu^2$$

$$= 1^2 \cdot \frac{1}{n} + 2^2 \cdot \frac{1}{n} + \dots + n^2 \cdot \frac{1}{n} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{1}{n} (1^2 + 2^2 + \dots + n^2) - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{n+1}{2} \left(\frac{2n+1}{3} - \frac{n+1}{2} \right)$$

$$= \frac{n+1}{2} \left(\frac{4n+2-3n-3}{6} \right) = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}$$

Q. A sample of 3 items is selected at random from a box containing 10 items of which 4 are defective. Find the expected number of defective items?

Sol:- wkt $\mu = np$ $n=3$ given
let P be the probability of having defective items, then $P = \frac{4}{10}$
 $\therefore E(X) = \mu = 3 \times \frac{4}{10} = \frac{12}{10} = 1.2$

(or)
Total no. of ways of drawing 3 from 10 =
Probability that there are no defective items $P(X=0) = \frac{{}^6C_3}{{}^{10}C_3} = \frac{1}{6}$

$$P(X=1) = \frac{{}^6C_2 \cdot {}^4C_1}{{}^{10}C_3} = \frac{15}{120} = \frac{1}{8}$$

$$P(X=2) = \frac{{}^6C_1 \cdot {}^4C_2}{{}^{10}C_3} = \frac{3}{120} = \frac{1}{40}$$

$$P(X=3) = \frac{{}^6C_0 \cdot {}^4C_3}{{}^{10}C_3} = \frac{1}{120} = \frac{1}{120}$$

Probability Distribution —

X	0	1	2	3
$P(X)$	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{3}{10}$	$\frac{1}{120}$

wkt. $E(x) = \sum_{i=0}^3 x_i p_i = (0)\left(\frac{1}{6}\right) + (1)\left(\frac{2}{10}\right) + 2\left(\frac{3}{10}\right) + 3\left(\frac{4}{10}\right) = \frac{12}{10} = 1.2$

Continuous Probability Distributions

Q. Given that $f(x) = \frac{k}{2^x}$, is a probability distribution for a random variable x that can take on the values $x = 0, 1, 2, 3$ and 4 .
(i) Find k . (ii) Mean and variance of x .

Sol: - Given $f(x) = \frac{k}{2^x}$, a probability distribution for a random variable x .

$$\therefore f(x) > 0 \quad \forall x$$

$$\text{and } \sum f(x) = 1 \quad \forall x$$

Given $x = 0, 1, 2, 3, 4$.

when $x = 0$, $f(x)$ is not defined

$\therefore$ we take $x = 1, 2, 3, 4$.

The probability distribution is given in the following table.

x	1	2	3	4
$f(x)$	$\frac{k}{2}$	$\frac{k}{4}$	$\frac{k}{6}$	$\frac{k}{8}$

(i) To find k :-

wkt $\frac{k}{2} + \frac{k}{4} + \frac{k}{6} + \frac{k}{8} = 1$

$$\rightarrow \frac{12k + 6k + 4k + 3k}{24} = 1$$

$$\rightarrow \frac{25k}{24} = 1 \rightarrow k = \frac{24}{25}$$

$$\text{Mean} = \sum_{i=1}^4 x_i f(x_i) = 1\left(\frac{k}{2}\right) + 2\left(\frac{k}{4}\right) + 3\left(\frac{k}{8}\right) + 4\left(\frac{k}{8}\right) = 2k = \frac{48}{25}$$

$$\text{Variance} = \frac{696}{625} = 1.1232$$

Q. For the following probability distribution

X	-3	6	9
P(X)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find (i) $E(X)$, (ii) $E(X^2)$, (iii) $E(2X+1)$

Sol (i) $E(X) = \sum_{i=1}^3 x_i p_i = (-3)\left(\frac{1}{6}\right) + 6\left(\frac{1}{2}\right) + 9\left(\frac{1}{3}\right)$
 $= \frac{11}{2}$

(ii) $E(X^2) = \sum_{i=1}^3 x_i^2 p_i = 9\left(\frac{1}{6}\right) + 36\left(\frac{1}{2}\right) + 81\left(\frac{1}{3}\right)$
 $= \frac{9}{6} + 18 + 27 = \frac{93}{2}$

(iii) $E[2X+1] = E[4X^2 + 4X + 1]$
 $= E(4X^2) + E(4X) + E(1)$
 $= 4E(X^2) + 4E(X) + 1$
 $= 4\left(\frac{93}{2}\right) + 4\left(\frac{11}{2}\right) + 1$
 $= 2(93) + 2(11) + 1 = 186 + 22 + 1 = 209$

Median: — The median, m , is the smallest value such that $P(X \leq m) \geq \frac{1}{2}$

Mode: — The mode is the value of the random Variable that has the maximum probability

Continuous Random Variable: - A Random variable x is said to be continuous if it can take any value in a given interval.

Ex: Time taken by a process.
Temperature
Length, weight, voltage.

Probability density function: - Let x be a continuous random variable. Then it is described by a Probability Density Function (PDF), $f(x)$, such that

(i) $f(x) \geq 0 \forall x$

(ii). $\int_{-\infty}^{\infty} f(x) dx = 1$

(iii). $P(a < x < b) = \int_a^b f(x) dx$

→ for continuous random variables,

$P(x=a) = 0$

Cumulative Distribution Function (CDF)

The CDF of x is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Properties

(i) $0 \leq F(x) \leq 1$

(ii) $F(x)$ is non-decreasing

(iii). $\lim_{x \rightarrow \infty} F(x) = 1$

$F(-\infty) = 0$ & $F(\infty) = 1$

$$(iv) F(-\infty) = 0$$

$$(v) F(\infty) = 1$$

$$(vi) P(a \leq x \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

Mean: - Mean of a distribution is given by $\mu = E(x) = \int_{-\infty}^{\infty} x f(x) dx$.

→ If x is defined from a to b , then

$$\mu = E(x) = \int_a^b x f(x) dx$$

Let x be a c.r.v. with PDF, $f(x)$ and CDF, $F(x)$. Then

Median: - The median m is the value such that $P(X \leq m) = \frac{1}{2}$

i.e. we solve $\int f(x) dx = \frac{1}{2}$ or

$$\int_m f(x) dx = \frac{1}{2} \text{ for } m.$$

i.e. set $F(m) = \frac{1}{2}$ & solve for m .

Mode: - The mode is the value of x at which the PDF, $f(x)$, is maximum.

∴ Mode is thus given by $f'(x) = 0$

and $f''(x) < 0$ for $a \leq x \leq b$.

Variance: - $\text{Var}(x) = \sigma^2 = E(x^2) - (E(x))^2$
 $= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$ or $\int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$

→ If x is defined from a to b , then

$$\sigma^2 = \int_a^b (x-\mu)^2 f(x) dx = \int_a^b x^2 f(x) dx - \mu^2$$

Q. The probability density, $f(x)$ of a continuous random variable is given by
 $f(x) = c e^{-|x|}$, $-\infty < x < \infty$. S.T. $c = 1/2$ and
 find the mean and variance of the distribution. Also find the probability that the variate lies between 0 and 4.

Sol: - Given $f(x) = c e^{-|x|}$, $-\infty < x < \infty$

(i) we have $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\rightarrow \int_{-\infty}^{\infty} c e^{-|x|} dx = 1 \rightarrow c \int_{-\infty}^{\infty} e^{-|x|} dx = 1$$

$$\rightarrow 2c \int_0^{\infty} e^{-x} dx = 1 \quad (\because e^{-|x|} \text{ is an even function})$$

$$\rightarrow 2c \int_0^{\infty} e^{-x} dx = 1 \quad (\because |x| = x \text{ in } [0, \infty])$$

$$\rightarrow 2c \left[-e^{-x} \right]_0^{\infty} = 1$$

$$\rightarrow -2c \left[e^{-\infty} - e^0 \right] = 1 \rightarrow -2c \left[0 - 1 \right] = 1$$

$$\rightarrow -2c(-1) = 1 \rightarrow 2c = 1 \rightarrow c = \frac{1}{2}$$

(ii) Mean = $\int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} x e^{-|x|} dx = 0$

$\because x e^{-|x|}$ is an odd function.

(let $g(x) = x e^{-|x|}$
 then $g(-x) = -x e^{-|-x|} = -x e^{-|x|} = -g(x)$

$\therefore g(x)$ is odd

(iii) variance of the distribution

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx = \mu^2$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x^2 e^{-|x|} dx \quad (\because \mu = 0, f(x) = \frac{1}{2} e^{-|x|})$$

$$= \frac{1}{2} \cdot 2 \int_0^{\infty} x^2 e^{-x} dx \quad (\because x^2 e^{-|x|} \text{ is even})$$

$$= \int_0^{\infty} x^2 \cdot e^{-x} dx \quad (\because |x| = x \text{ in } (0, \infty))$$

$$= \left[x^2 \frac{e^{-x}}{-1} - 2x(e^{-x}) + 2 \frac{e^{-x}}{-1} \right]_0^{\infty}$$

$$= (0 - 0 + 0) - (0 - 0 - 2e^0) = 2$$

(iv) The probability between 0 and 4:-

$$\begin{aligned}
 P(0 \leq x \leq 4) &= \frac{1}{2} \int_0^4 e^{-x} dx = \frac{1}{2} \left[-e^{-x} \right]_0^4 \\
 &= \frac{1}{2} \left[-e^{-4} + 1 \right] = \frac{1 - e^{-4}}{2}
 \end{aligned}$$

Q. PDF of a random variable X is ~~for~~ $\frac{1}{2} \sin x$

$$f(x) = \begin{cases} \frac{1}{2} \sin x, & \text{for } 0 \leq x \leq \pi \\ 0, & \text{otherwise.} \end{cases}$$

Find the mean, mode and median of the distribution and also find the probability between 0 and $\pi/2$.

Sol:— Given $f(x) = \begin{cases} \frac{1}{2} \sin x, & \text{for } 0 \leq x \leq \pi \\ 0, & \text{otherwise} \end{cases}$ — (1)

(i) Mean = $\int_{-\infty}^{\infty} x f(x) dx$

$$= \int_{-\infty}^{\infty} x f(x) dx + \int_0^{\pi} x f(x) dx + \int_{\pi}^{\infty} x f(x) dx$$

$$= 0 + \int_0^{\pi} x f(x) dx + 0, \text{ from (1)}$$

$$= \int_0^{\pi} x \cdot \frac{1}{2} \sin x dx = \frac{1}{2} \int_0^{\pi} x \sin x dx$$

$$= \frac{1}{2} \left[-x \cos x - (1)(-\sin x) \right]_0^{\pi}$$

$$= \frac{1}{2} \left[-x \cos x + \sin x \right]_0^{\pi}$$

$$= \frac{1}{2} \left[-\pi \cos \pi \right] = -\frac{\pi}{2} (-1) = \frac{\pi}{2}$$

$$\because \cos \pi = -1, \sin 0 = \sin \pi = 0$$

(ii). Mode is the max^{im} value of x for which $f(x)$ is maximum

$$f(x) = \frac{1}{2} \sin x$$

for maximum, $f'(x) = 0$.

$$\Rightarrow \frac{1}{2} \cos x = 0 \rightarrow \cos x = 0$$

$$\Rightarrow x = \frac{\pi}{2}$$

and at maximum, $f''(x) < 0$

$$f''(x) = -\frac{1}{2} \sin x$$

$$f''\left(\frac{\pi}{2}\right) = -\frac{1}{2} \sin \frac{\pi}{2} = -\frac{1}{2} < 0$$

hence $f(x)$ is maximum at $x = \frac{\pi}{2}$

$\therefore$ Mode of the distribution is $x = \frac{\pi}{2}$

(ii). If M is the median of the distribution, then $\int_{-\infty}^M f(x) dx = \int_M^{\infty} f(x) dx = \frac{1}{2}$

$$\Rightarrow \int_0^M \frac{1}{2} \sin x dx = \frac{1}{2}$$

$$\Rightarrow \left[-\frac{1}{2} \cos x \right]_0^M = \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} [\cos M - \cos 0] = \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} [\cos M - 1] = \frac{1}{2}$$

$$\Rightarrow [1 - \cos M] = 1$$

$$\Rightarrow \cos M = 0 \rightarrow M = \frac{\pi}{2}$$

hence the median = $\frac{\pi}{2}$

Thus Mean = Median = Mode = $\frac{\pi}{2}$

$$\begin{aligned}
 (iv) P(0 < x < \frac{\pi}{2}) &= \int_0^{\pi/2} f(x) dx \\
 &= \int_0^{\pi/2} \frac{1}{2} \sin x dx = \frac{1}{2} [-\cos x]_0^{\pi/2} \\
 &= -\frac{1}{2} [\cos \frac{\pi}{2} - \cos 0] = -\frac{1}{2} [0 - 1] = \frac{1}{2}
 \end{aligned}$$

Q. For the continuous function $f(x) = kx^2 e^{-x}$ when $x \geq 0$, find (i) k (ii) Mean (iii) variance.

Sol (i) we know that $\int_0^{\infty} f(x) dx = 1$.

$$\rightarrow \int_0^{\infty} kx^2 e^{-x} dx = 1, \quad x \geq 0$$

$$\rightarrow \int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$\rightarrow k \left[x^2 \frac{e^{-x}}{-1} - \frac{2x}{-1} \frac{e^{-x}}{-1} + \frac{2}{-1} \frac{e^{-x}}{-1} \right]_0^{\infty} = 1$$

$$\rightarrow k \left[-x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \right]_0^{\infty} = 1$$

$$\rightarrow k \left[0 - 2 \right] = 1 \rightarrow 2k = 1 \rightarrow \boxed{k = \frac{1}{2}}$$

$$(ii) \text{ Mean} = \int_0^{\infty} x f(x) dx = k \int_0^{\infty} x^3 e^{-x} dx$$

$$= k \left[x^3 \frac{e^{-x}}{-1} - \frac{3x^2}{-1} \frac{e^{-x}}{-1} + \frac{6x}{-1} \frac{e^{-x}}{-1} - \frac{6}{-1} \frac{e^{-x}}{-1} \right]_0^{\infty}$$

$$= k \left[(0 + 6) \right] = \frac{1}{2} (6) = 3$$

$$\left[0 - \frac{6}{1} \right] = -6$$

$$= 6$$

$$(iii) - \text{Variance} = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

$$= \int_{-\infty}^{\infty} x^2 k x^2 e^{-x} dx - 3^2$$

$$= k \int_{-\infty}^{\infty} x^4 e^{-x} dx - 9$$

$$= k \left[\frac{x^4 e^{-x}}{-1} - \frac{4x^3 e^{-x}}{+1} + \frac{12x^2 e^{-x}}{-1} - \frac{24x e^{-x}}{+1} + \frac{24 e^{-x}}{-1} \right]_{-\infty}^{\infty}$$

$$= k [0 - (-24)] - 9 = 24k - 9$$

$$0 \leq x < 1 \Rightarrow 24(3) - 9$$

$$= 72 - 9 = 63$$

A CRV x has the PDF, $f(x) = \begin{cases} kx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ find (i) $P(x)$

Q. Given $f(x) = kx^2, 0 \leq x \leq 1$. find k .

Sol f. (i) wkt $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^{\infty} kx^2 dx = 1$$

$$\Rightarrow k \left[\frac{x^3}{3} \right]_0^1 = 1 \Rightarrow \frac{k}{3} [1 - 0] = 1 \Rightarrow k = 3$$

$$(ii) \cdot P(x < 0.5) = \int_{-\infty}^{0.5} 3x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_0^{0.5}$$

$$= \frac{3}{3} [(0.5)^3 - 0]$$

$$= (0.5)^3 = 0.125$$

Q. The PDF of a random variable, x is given by $f(x) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$
 find 1. The cumulative distribution fn (CDF)
 2. $P(1 < x < 3)$.

Sol: — Given $f(x) = \frac{1}{4}, 0 \leq x \leq 4$.

(i) CDF :- $F(x) = \int_0^x f(t) dt$
 $\Rightarrow \int_0^x \frac{1}{4} dt = \left[\frac{t}{4} \right]_0^x = \frac{x}{4}$

$\therefore f(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{4}, & 0 \leq x \leq 4 \\ 1, & x > 4 \end{cases}$

Q. A CRV X has the distribution fn,

$$F(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ k(x-1)^4, & \text{if } 1 < x \leq 3 \\ 1, & \text{if } x > 3. \end{cases}$$

Determine (i) $f(x)$ (ii) k (iii) Mean

Sol

wkt $F'(x) = f(x)$

$$\therefore f(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ 4k(x-1)^3, & \text{if } 1 < x \leq 3 \\ 0 & \text{if } x > 3 \end{cases}$$

$$\text{wkt } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\rightarrow \int_{-\infty}^1 f(x) dx + \int_1^3 f(x) dx + \int_3^{\infty} f(x) dx = 1$$

$$\rightarrow \int_1^3 f(x) dx = 1$$

$$\rightarrow \int_1^3 4k(x-1)^3 dx = 1$$

$$\rightarrow \left[k4(x-1)^4 \right]_1^3 = 1$$

$$\rightarrow \left[4k \left[8 \right] = 1 \rightarrow k = \frac{1}{32} \right]$$

$$4k \left[\frac{(x-1)^4}{4} \right]_1^3 = 1$$

$$\rightarrow k [16] = 1 \rightarrow k = \frac{1}{16}$$

$$(iii) \text{ Mean} = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^3 x \cdot \frac{1}{4} (x-1)^3 dx$$

$$= \frac{1}{4} \int_0^3 x (x^3 - 3x^2 + 3x - 1) dx$$

$$= \frac{1}{4} \int_0^3 (x^4 - 3x^3 + 3x^2 - x) dx$$

$$= \frac{1}{4} \left[\frac{x^5}{5} - \frac{3x^4}{4} + \frac{3x^3}{2} - \frac{x^2}{2} \right]_0^3$$

$$= \frac{1}{4} \left[\frac{3^5}{5} - \frac{3 \cdot 3^4}{4} + \frac{27 \cdot 3}{2} - \frac{1}{2} \right]$$

$$= \frac{1}{4} \left[\frac{243}{5} - \frac{243}{4} + \frac{81}{2} + 20 \right]$$

$$= \frac{1}{4} \left[\frac{242}{5} - \frac{240}{4} + 20 \right]$$

$$= \frac{1}{4} \left[\frac{968 - 1200}{20} + 20 \right]$$

$$= \frac{1}{4} \left[\frac{502}{20} + 20 \right]$$

$$= \frac{1}{4} [25.1 + 20] = \frac{45.1}{4}$$

$$= 2.6$$

Equally Likely Events:— Two or more events are said to be equally likely if each event has the same chance of occurring.
i.e. when there is no reason to expect one event to occur more than the others under the given conditions.

Ex: when a fair die is rolled, getting 1, 2, 3, 4, 5 or 6 are equally likely events.

Exhaustive Events:— Events are said to be exhaustive if their union contains all possible outcomes of the experiment.

Ex: when a die is thrown, the events $\{1, 2\}$, $\{3, 4, 5\}$, $\{6\}$ are exhaustive since their union contains all possible outcomes.

Mutually Exclusive Events:— Events are mutually exclusive if the occurrence of one event excludes the occurrence of the other.

Ex: when a die is rolled, the events getting an even number and getting an odd number are mutually exclusive.